Sticky priors and the most likely count vector

Working note S3 (moved out of Paper C, October 2026)

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1 Setting

This note was Section 6 and Appendix D of an earlier version of Paper C, "Face selection for slowly switching Markov chains" [5]. When Paper B [4] was merged into Paper C in October 2026, the application to sticky priors moved here. The mathematics is unchanged. The statement numbers of [5] that we quote are those of the merged paper.

We recall what we need from [5]. A chain on d states has the transition matrix $I + hQ$, where Q is a generator with rates $q_{ij} \geq 0$ for $i \neq j$ and exit rates $q_i = \sum_{j \neq i} q_{ij}$. It runs for N steps, its first state has the law μ, and $h = T/N$, $L = \log N$ and $\tau = T/L$. The composition n counts the visits to each state, and $P_N(n)$ is its law. A face is a nonempty set F of states, its exit rate λ_F is minus the Perron root of Q restricted to F, and for an irreducible face $\alpha_F^* = l_F \circ r_F$ is its quasi-ergodic law, the entrywise product of the left and right Perron vectors with $l_F \cdot r_F = 1$. Put $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$. First-order face selection (Theorem 3.3 of [5]) says that for large N every most likely composition has a support that minimizes Φ_τ among the faces that a path can use, and that its fractions are within $O(1/L)$ of α_F^* when these faces are irreducible.

For two states with $q_{12} = q_{21} = 1$, $S_{a,b}(u)$ is the sum of $u^{s(w)}$ over the words w with a letters 1 and b letters 2, where $s(w)$ is the number of switches of w, and $S_N = S_{\lfloor N/2 \rfloor, \lceil N/2 \rceil}$. The equation $S_N(u) = 1$ has exactly one positive root u_N. Put $p_N = 1/(1 + u_N)$, $z_N = N(1 - p_N)$ and $c_0 = \frac{1}{2}\log(\pi/8)$. Paper A [3] studies these numbers at every N.

2 Sticky priors and the most likely count vector

In a hidden Markov model the hidden states $X_1, \ldots, X_N$ form a Markov chain on d states. Before any data are seen, the model already says how the N time steps are likely to be shared among the states. This is the law P_N of the count vector $n = (n_1, \ldots, n_d)$, where n_i is the number of time steps in state i (the composition of [5]). A sticky prior makes the chain leave its current state rarely, which is the regime of [5]. We ask which states the most likely count vector uses.

Here is what we use from [5]. Suppose the chain leaves its state with probability $h = \tau \log N/N$ at each step and then picks the next state from a kernel Π. For a set F of states (a face) let ρ_F be the Perron root of Π restricted to F. The chain stays in F for all N steps with probability about $N^{-\tau(1-\rho_F)}$, and about $N^{|F|-1}$ count vectors use exactly the states of F. Theorem 3.3 of [5] says that for large N the most likely count vector uses a set F that minimizes $\Phi_\tau(F) = \tau(1 - \rho_F) + |F| - 1$, and that its fractions n/N are then close to the law α_F^*, the entrywise product of the left and right Perron vectors of Π on F. Besides this we use the description of a tie between several minimizers

(Theorem 4.8 of [5]) and the exact identity for a symmetric pair of states (Proposition 5.1 of [5]), once each.

The answer is as follows. As the stickiness decreases, that is, as τ grows, the most likely count vector moves from one state to larger sets of states. Among symmetric kernels only the uniform one jumps from one state straight to all of them.

Fox et al. [2] add a self-transition bias $\kappa \geq 0$ to the HDP hidden Markov model (HDP stands for hierarchical Dirichlet process), so that the hidden chain stays in each state longer. In their weak-limit approximation with base weights $\bar{\beta}$ and concentration ω, row j of the transition matrix $(\tilde{\pi}_{jl})$ is Dirichlet with parameters $\omega\bar{\beta}_1, \dots, \omega\bar{\beta}_j + \kappa, \dots, \omega\bar{\beta}_d$. By neutrality, row j off the diagonal, divided by $1 - \tilde{\pi}_{jj}$, is Dirichlet with parameters $(\omega\bar{\beta}_l)_{l \neq j}$ and independent of $\tilde{\pi}_{jj}$. So κ only sets how sticky the chain is, and the kernel of the switches does not depend on it. In this section we fix every exit probability $1 - \tilde{\pi}_{jj}$ at the same value $h = \tau L/N$, the scale of the crossing of the two-state walk, and study the kernel. Under the prior the exit probabilities are random. By the aggregation property of the Dirichlet law, $1 - \tilde{\pi}_{jj}$ has the law $\mathrm{Beta}(\omega(1 - \bar{\beta}_j), \omega\bar{\beta}_j + \kappa)$, with mean $\omega(1 - \bar{\beta}_j)/(\omega + \kappa)$, so a mean of order L/N needs $\kappa \asymp N/L$. Mixing over this law favors single states. A single state then keeps probability $\mu_j \mathbb{E}[\tilde{\pi}_{jj}^{N-1}]$, which decays only like a power of L when $\kappa \asymp N/L$, while at a fixed h of order L/N it decays like a power of N. So the mode of the mixture can differ from the mode studied here, and we do not treat it. All results of this section are about the law given Π and h.

So let $\Pi = (\pi_{ij})$ be a stochastic matrix on $[d]$ with zero diagonal, $\tau > 0$ and $h = \tau L/N < 1$. The hidden chain has transition matrix $(1 - h)I + h\Pi = I + hQ$ with $Q = \Pi - I$, so $q_i = 1$ and $T = \tau L$ (Example 2.5(iii) of [5]). We assume that $\pi_{ij} > 0$ for all $i \neq j$, which we call (A+), and that the start μ has full support. Under the prior, (A+) holds almost surely. Let $\rho_F = \varrho(\Pi_F)$ for the principal submatrix Π_F on F. Then the exit rate of F is $\lambda_F = 1 - \rho_F$, and Q_F and Π_F have the same Perron vectors. A vertex (a single state) has $\rho_F = 0$, and $\rho_{[d]} = 1$. If $2 \leq |F| \leq d - 1$, then Π_F is irreducible by (A+), and $0 < \rho_F < 1$ since ρ_F is at least the least row sum of Π_F and the Perron root of the irreducible Π exceeds that of each proper principal submatrix [1]. So $\Phi_\tau(F) = \tau(1 - \rho_F) + |F| - 1$, and we put

$$\tau_- = \min_{|F| \geq 2} \frac{|F| - 1}{\rho_F}, \qquad \tau_+ = \max_{F \neq [d]} \frac{d - |F|}{1 - \rho_F}.$$

We call a global mode of P_N, that is, a most likely value of the count vector before any data are seen, given Π and h, a *conditional mode*.

Corollary 2.1 (support and location of the conditional mode)**.** *Assume* (A+)*, let μ have full support, and fix $\tau > 0$. Let $\mathcal{F}_\tau$ be the set of faces that minimize Φ_τ. There is $N_0(\tau)$ such that for $N \geq N_0(\tau)$ every conditional mode n^* satisfies $\operatorname{supp} n^* \in \mathcal{F}_\tau$ and $n^*/N = \alpha_F^* + O(1/L)$ with $F = \operatorname{supp} n^*$. If $\mathcal{F}_\tau = \{F^*\}$, then $n^*/N \to \alpha_{F^*}^* = l \circ r$, where l and r are the left and right Perron vectors of Π_{F^*} with $l \cdot r = 1$. No reversibility is needed.*

In words, for large N a conditional mode uses exactly the states of a set that minimizes Φ_τ, and it shares the N time steps among them in the proportions α_F^*.

Proof. By (A+) every face is irreducible and, as μ has full support, admissible. So Theorem 3.3(iii) and (iv) of [5] give the claims, with $\alpha_F^* = l \circ r$. □

The threshold $N_0(\tau)$ is not uniform in τ. At $\tau = \tau_-$ the vertices tie with the faces that attain τ_-, and Theorem 4.8(iii) of [5] with $\tau^* = \tau_-$ and a fixed large $t = t_0$ shows that for large N the conditional mode is not a single state at $\tau = \tau_-(1 - \frac{\log L - 2t_0}{2L}) < \tau_-$. So $N_0(\tau) \to \infty$ as $\tau \uparrow \tau_-$.

The next proposition says for which τ the conditional mode uses one state, all states, or a set in between.

Proposition 2.2 (thresholds)**.** *Assume* (A+) *and* $d \geq 3$*.*

(a) The minimizers of Φ_τ *are exactly the vertices if and only if* $\tau < \tau_-$*, and no vertex minimizes* Φ_τ *if* $\tau > \tau_-$*. The face* $[d]$ *is the unique minimizer if and only if* $\tau > \tau_+$*, and* $[d]$ *does not minimize* Φ_τ *if* $\tau < \tau_+$*.*

(b) The thresholds satisfy $1 < \tau_- \leq d-1 \leq \tau_+$*.*

(c) The equality $\tau_- = d-1$ *holds if and only if* $\tau_+ = d-1$*, and both hold if and only if*

$$\rho_F \leq \frac{|F|-1}{d-1} \qquad \text{for every face } F. \tag{1}$$

If (1) *fails, then* $\tau_- < d-1 < \tau_+$*, and for* $\tau \in (\tau_-, \tau_+)$ *every minimizer* F *of* Φ_τ *has* $2 \leq |F| \leq d-1$*.*

Consequently, let μ *have full support, fix* τ *and let* $N \geq N_0(\tau)$ *as in Corollary 2.1. If* $\tau < \tau_-$*, the prior modes are the* Ne_i *with* $\mu_i = \max_l \mu_l$*. If* $\tau > \tau_+$*, every conditional mode has full support. If* $\tau_- < \tau < \tau_+$*, every prior mode has a support* F *with* $2 \leq |F| \leq d-1$*, and* $F = F^*$ *if* F^* *is the unique minimizer of* Φ_τ*.*

Proof. We compare every face with a vertex and with $[d]$. The function $\Phi_\tau(F) = \tau(1-\rho_F) + |F| - 1$ is affine in τ, so each comparison changes sign at one value of τ, and τ_- and τ_+ are the extreme ones.

(a) A vertex has $\Phi_\tau = \tau$. For $|F| \geq 2$ we have $\rho_F > 0$ and

$$\Phi_\tau(F) - \tau = |F| - 1 - \tau\rho_F = \rho_F\Big(\frac{|F|-1}{\rho_F} - \tau\Big).$$

So F is worse than a vertex exactly when $\tau < (|F|-1)/\rho_F$. This holds for every F with $|F| \geq 2$ exactly when $\tau < \tau_-$, and if $\tau > \tau_-$ some F is better than every vertex. In the same way $\Phi_\tau([d]) = d-1$, and for $F \neq [d]$ we have $1 - \rho_F > 0$ and

$$\Phi_\tau(F) - (d-1) = \tau(1-\rho_F) - (d - |F|) = (1-\rho_F)\Big(\tau - \frac{d-|F|}{1-\rho_F}\Big).$$

So F is worse than $[d]$ exactly when $\tau > (d-|F|)/(1-\rho_F)$, which gives the two claims on $[d]$.

(b) In the minimum that defines τ_-, a face with $2 \leq |F| \leq d-1$ has $\rho_F < 1$ and so $(|F|-1)/\rho_F > 1$, and $[d]$ gives the value $d-1 \geq 2$. So $\tau_- > 1$ and $\tau_- \leq d-1$. In the maximum that defines τ_+, a vertex gives the value $d-1$, so $\tau_+ \geq d-1$.

(c) By (b), $\tau_- = d-1$ exactly when $(|F|-1)/\rho_F \geq d-1$ for all $|F| \geq 2$, and $\tau_+ = d-1$ exactly when $(d-|F|)/(1-\rho_F) \leq d-1$ for all $F \neq [d]$. After clearing the denominators both conditions say $(d-1)\rho_F \leq |F|-1$. This inequality always holds for the vertices and for $[d]$, so each condition is (1). If (1) fails, then neither holds, so $\tau_- < d-1 < \tau_+$ by (b), and (a) excludes the vertices and $[d]$ on (τ_-, τ_+).

For the consequence, Corollary 2.1 says that the support of a prior mode minimizes Φ_τ, and (a) and (c) say which faces can do so. If $\tau < \tau_-$ the conditional modes are vertex compositions, and $P_N(Ne_i) = \mu_i(1-h)^{N-1}$ is largest for the largest μ_i. Parts (a) to (c) are checked in Lean. □

We say that Π has a *direct jump* when (1) holds. Then at first order the conditional mode goes from one state straight to all d states at $\tau = d-1$. For the uniform kernel $\pi_{ij} = 1/(d-1)$ the matrix Π_F has constant row sums, so $\rho_F = (|F|-1)/(d-1)$. Hence (1) says that every set of states is at least as easy to leave as under the uniform kernel. That kernel has $Q = (J - dI)/(d-1)$, and all its faces tie at $\tau = d-1$. This is the degeneracy of the complete graph, the Potts chain of [4], which [5, Section 5] treats. The next corollary shows that a direct jump is special.

Corollary 2.3 (only the uniform kernel)**.** *Let $d \geq 3$ and assume* (A+). (a) *If Π is symmetric, then Π has a direct jump if and only if $\pi_{ij} = 1/(d-1)$ for all $i \neq j$.* (b) *If $d = 3$, then Π has a direct jump if and only if $\pi_{ij}\pi_{ji} \leq \frac{1}{4}$ for all three pairs. In particular every cyclic kernel $\pi_{i,i+1} = a$, $\pi_{i,i-1} = 1-a$ (indices mod 3) has a direct jump.*

Proof. We test (1) on the pairs. The matrix $\Pi_{\{i,j\}}$ has zero diagonal and off-diagonal entries π_{ij} and π_{ji}, so $\rho_{\{i,j\}} = \sqrt{\pi_{ij}\pi_{ji}}$.

(a) Here $\rho_{\{i,j\}} = \pi_{ij}$, so for a pair (1) reads $\pi_{ij} \leq 1/(d-1)$. The d rows sum to 1 and have $d(d-1)$ off-diagonal entries, so the average of the π_{ij} is $1/(d-1)$. If none of them exceeds the average, they are all equal to it. So a direct jump forces the uniform kernel. Conversely the uniform kernel has $\rho_F = (|F|-1)/(d-1)$ for every face, as we saw above, so it satisfies (1) with equality.

(b) For $d = 3$ the faces are the vertices, the three pairs and [3]. For the vertices and [3], (1) reads $0 \leq 0$ and $1 \leq 1$. For a pair it reads $\sqrt{\pi_{ij}\pi_{ji}} \leq \frac{1}{2}$. The cyclic kernel has $\pi_{ij}\pi_{ji} = a(1-a) \leq \frac{1}{4}$ for every pair. Both parts are checked in Lean. □

So for a symmetric kernel that is not uniform there is a range of τ in which the conditional mode uses more than one state but not all of them (Proposition 2.2(c)). Without symmetry a direct jump is possible, and for three states we can say how likely it is under the prior.

Proposition 2.4 (the direct jump under the prior)**.** *Let $d = 3$ and let the base weights be uniform. In the weak-limit sticky HDP-HMM the rows of Π are then independent, and the first off-diagonal entry of each row has the law* Beta$(\omega/3, \omega/3)$. *Write $\mathbb{P}_\omega$ for this law of Π and D for the event that Π has a direct jump. Then* (a) $\mathbb{P}_\omega(D) \to \frac{1}{4}$ *as $\omega \to 0$, and* (b) $\mathbb{P}_\omega(D) = \frac{3\sqrt{3}}{8\pi\omega}(1 + o(1))$ *as $\omega \to \infty$.*

Proof. Let B_1, B_2, B_3 be independent with law Beta$(\omega/3, \omega/3)$, and put $\pi_{12} = B_1$, $\pi_{21} = B_2$ and $\pi_{31} = B_3$, so that the other off-diagonal entries are $1 - B_j$. By Corollary 2.3(b), D is the event that B_1B_2, $(1-B_1)B_3$ and $(1-B_2)(1-B_3)$ are all at most $\frac{1}{4}$.

(a) As $\omega \to 0$ each B_j tends in law to a fair coin on $\{0,1\}$. So the law of (B_1, B_2, B_3) tends to the uniform law on the eight points of $\{0,1\}^3$. At these points the three products take only the values 0 and 1, so none of the points lies on the boundary of D (where some product equals $\frac{1}{4}$). Hence $\mathbb{P}_\omega(D)$ tends to the limit probability of D, which is the fraction of the eight points at which all three products are 0. Only $(1,0,1)$ and $(0,1,0)$ have this property (checked in Lean), so the limit is $\frac{2}{8} = \frac{1}{4}$. Part (b) is proved in Appendix A. □

In words, a direct jump has probability close to $\frac{1}{4}$ when the concentration is small and of order $1/\omega$ when it is large. For (b), for large ω the set D is a thin tube around the cyclic kernels, with cross-section of area about $\frac{9}{8}t^4$ at the logit coordinate t along the cyclic kernels (Appendix A), which gives the order $1/\omega$. So the uniform tie is not typical. At $\omega = 1$, Monte Carlo with 10^6 draws gives $\mathbb{P}_\omega(D) = 0.115$. So at $\omega = 1$ about 88 percent of the kernels drawn from the prior have a range of τ in which, at first order, the conditional mode sits on a pair (Proposition 2.2(c)).

The comparison of a single state with a symmetric pair is exact at every N.

Corollary 2.5 (an exact link to the two-state walk)**.** *Let $i \neq j$ satisfy $\pi_{ij} = \pi_{ji} = \pi_0 > 0$ and $\mu_i = \mu_j > 0$. We do not assume* (A+)*. Then for every $N \geq 2$ the balanced compositions are the most likely ones with support $\{i, j\}$, and they are more likely than Ne_i if and only if $\tau > \tau^{\mathrm{pair}}(N) := Nu_N/((\pi_0 + u_N)L)$. If also $\mu_i = \mu_j = \max_l \mu_l$, then for every τ with $\tau^{\mathrm{pair}}(N) < \tau < N/L$ no conditional mode is a single state. Finally,*

$$\tau^{\mathrm{pair}}(N) = \frac{1}{\pi_0}\Big[1 - \frac{\frac{1}{2}\log L - c_0}{L} + O\Big(\frac{\log L}{L^2}\Big)\Big],$$

which tends to the first-order value $1/\rho_{\{i,j\}} = 1/\pi_0$.

In words, a single state against a symmetric pair is the two-state walk, so the value of τ at which the pair overtakes the state is known at every N and not only in the limit.

Proof. We apply Proposition 5.1 of [5]. Every exit rate is 1, so its hypotheses hold with $a = \pi_0$ and $q = 1$, and its odds are $u = \pi_0 h/(1-h)$. It says that the balanced compositions are the most likely ones on $\{i, j\}$ and that they beat Ne_i exactly when $u > u_N$. Solving for h, this is $h > u_N/(\pi_0 + u_N)$, and with $h = \tau L/N$ it is $\tau > \tau^{\mathrm{pair}}(N)$.

For the conditional modes, $P_N(Ne_l) = \mu_l(1-h)^{N-1} \leq P_N(Ne_i)$ for every l. So when the pair beats Ne_i it beats every single state. The condition $\tau < N/L$ is $h < 1$, and the range of τ is not empty as $u_N/(\pi_0 + u_N) < 1$.

Finally $Nu_N = z_N/p_N$ and $u_N = O(L/N)$, so $\tau^{\mathrm{pair}}(N) = \frac{z_N}{\pi_0 L}(1 + O(L/N))$. By Corollary 5.2 of [5], $z_N + \frac{1}{2}\log z_N = L + c_0 + O(1/L)$ and z_N is of order L, so $\log z_N = \log L + O(\log L/L)$ and $z_N = L - \frac{1}{2}\log L + c_0 + O(\log L/L)$. Dividing by $\pi_0 L$ gives the claim. □

So far we looked at the single most likely count vector. The last result gives the total probability that the chain uses only the states of a set F.

Proposition 2.6 (face masses)**.** *Assume* (A+) *and $0 < h < 1$. For every face F with $|F| \geq 2$ and $\Lambda_F = 1 - h\lambda_F$,*

$$\frac{\min r_F}{\max r_F}\mu(F)\Lambda_F^{N-1} \leq \mathbb{P}(\operatorname{supp} n \subseteq F) \leq \frac{\max r_F}{\min r_F}\mu(F)\Lambda_F^{N-1},$$

and for fixed τ, $\Lambda_F^{N-1} = N^{-\tau\lambda_F}e^{O(L^2/N)}$. Also $\mathbb{P}(\text{one state}) = (1-h)^{N-1} \sim N^{-\tau}$ and $\mathbb{P}(\operatorname{supp} n = [d]) \to 1$ for fixed τ.

In words, the chain stays inside F with probability $\mu(F)N^{-\tau\lambda_F}$ up to a bounded factor, and the probability that it uses every state tends to 1.

Proof. We write the probability with the transition matrix restricted to F and compare the vector $\mathbf{1}$ with the Perron vector. Let $U_F = (1-h)I + h\Pi_F \geq 0$. The chain stays in F for all N steps with probability $\mathbb{P}(\operatorname{supp} n \subseteq F) = \mu_F^\top U_F^{N-1}\mathbf{1}$, and $U_F r_F = \Lambda_F r_F$. As $r_F/\max r_F \leq \mathbf{1} \leq r_F/\min r_F$ entrywise and U_F^{N-1} has nonnegative entries, we may replace $\mathbf{1}$ by these two multiples of r_F. This gives the bounds. For fixed τ, $(N-1)\log(1-h\lambda_F) = -\tau\lambda_F L + O(L^2/N)$. A vertex composition is a single path that never switches, which gives $\mathbb{P}(\text{one state})$. Every $F \neq [d]$ has $\lambda_F > 0$, so each of the finitely many proper faces has probability tending to 0, and $\mathbb{P}(\operatorname{supp} n = [d]) \to 1$. The finite bounds are checked in Lean. □

Remark 2.7 (conditional mode versus mass). Let $d \geq 3$. For fixed $\tau < \tau_+$ and large N every conditional mode has a proper support, while $\mathbb{P}(\operatorname{supp} n = [d]) \to 1$. For $\tau < \tau_-$ the conditional mode uses one state, an event of probability about $N^{-\tau}$. A face of size k holds about N^{k-1}

τ	conditional mode	one state	$\{1,2\}$	$[3]$
0.5	one state	0.0643	0.465	0.379
0.8	one state	0.0122	0.423	0.531
3	$(120,120,0)$	4.3×10^{-8}	0.129	0.871
8	$(111,111,18)$	1.2×10^{-21}	0.0081	0.992

Table 1: The kernel Π with rows $(0,0.9,0.1)$, $(0.9,0,0.1)$, $(0.5,0.5,0)$, the uniform start and $N=240$, where $\tau_-=10/9$ and $\tau_+=10$. The conditional mode (from exact dynamic programming, with a tie of three compositions at $\tau=0.5$ and 0.8) and the probabilities of the supports.

compositions, so it can carry most of the mass while each of them is less likely than a vertex. Table 1 shows this. At $\tau=0.8$ the conditional mode is a single state, yet $\mathbb{P}(\text{one state})=0.0122$. At $\tau=8$ first order puts the conditional mode on $\{1,2\}$, but at $N=240$ it has full support. So the conditional mode can be a poor summary of a sticky prior, and face selection tells us on which states it sits.

A The large-concentration limit of the direct jump

We prove Proposition 2.4(b), with B_1,B_2,B_3 independent $\mathrm{Beta}(a,a)$ as in part (a) and $a=\omega/3\geq 1$.

Idea of the proof. We pass to the logits ℓ_j of the B_j. In these coordinates the three conditions that define D say that three linear forms M_1,M_2,M_3 of ℓ, whose sum is 0, are bounded above by quantities of second order in ℓ. For large a the law of ℓ concentrates near 0 on the scale $a^{-1/2}$. There the bounds are small, so on D all three forms are nearly 0. The points where they vanish form the line of the cyclic kernels, so D is a thin tube around this line. Let t be the coordinate along the line. At the point t of the line each of the three bounds is about $\frac{1}{2}t^2$, so the section of D at t is close to the triangle $M_1,M_2,M_3\leq\frac{1}{2}t^2$, which has area $\frac{9}{8}t^4$. So we expect $\mathbb{P}_\omega(D)\approx a^{3/2}\int t^4e^{-cat^2}dt\approx 1/a$, and a Gaussian integral gives the constant.

Before the proof we set up three things, namely the logits and the form of D in them, two bounds on the function that appears, and coordinates along and across the cyclic kernels.

Logits. Put $\ell_j=\log(B_j/(1-B_j))$, $g(y)=2\log\cosh(y/2)$ and $g_j=g(\ell_j)$. Since $B_j=e^{\ell_j/2}/(2\cosh(\ell_j/2))$ and $1-B_j=e^{-\ell_j/2}/(2\cosh(\ell_j/2))$, we have

$$\log(2B_j)=\tfrac{1}{2}(\ell_j-g_j),\qquad \log\bigl(2(1-B_j)\bigr)=-\tfrac{1}{2}(\ell_j+g_j),\qquad B_j(1-B_j)=\tfrac{1}{4}e^{-g_j}.$$

The function g is even and convex with $g(0)=g'(0)=0$, so $g\geq 0$. The density of B_j is proportional to $(B_j(1-B_j))^{a-1}$, and $dB_j/d\ell_j=B_j(1-B_j)$. So ℓ_j has the density $e^{-ag(\ell_j)}/Z$ with $Z=4^a\Gamma(a)^2/\Gamma(2a)=2\sqrt{\pi}\,\Gamma(a)/\Gamma(a+\frac{1}{2})$, where the second form is the duplication formula for Γ.

By the proof of Proposition 2.4, D is the event that B_1B_2, $(1-B_1)B_3$ and $(1-B_2)(1-B_3)$ are all at most $\frac{1}{4}$. We multiply each product by 4 and take logarithms with the display above. For example the first condition becomes $\frac{1}{2}(\ell_1-g_1)+\frac{1}{2}(\ell_2-g_2)\leq 0$. So D is the set where

$$M_1\leq g_1+g_2,\qquad M_2\leq g_1+g_3,\qquad M_3\leq g_2+g_3,\tag{2}$$

with $M_1=\ell_1+\ell_2$, $M_2=\ell_3-\ell_1$ and $M_3=-\ell_2-\ell_3=-M_1-M_2$. Let $\psi=g_1+g_2+g_3$ (in this appendix ψ is not the Bessel function of [5]). Then the joint density of $\ell=(\ell_1,\ell_2,\ell_3)$ is $Z^{-3}e^{-a\psi(\ell)}$, and

$$\mathbb{P}_\omega(D)=Z^{-3}\int_D e^{-a\psi(\ell)}d\ell.\tag{3}$$

Two bounds on g. For real y,

$$\frac{y^2}{4} - \frac{y^4}{96} \le g(y) \le \frac{y^2}{4}, \tag{g1}$$

$$g(y) \ge \frac{1}{10}\min(y^2, |y|). \tag{g2}$$

The first says that $g(y)$ is close to $y^2/4$ near 0, and we use it near the cyclic kernels. The second is a rough lower bound that we use far from 0. For (g1) put $x = y/2$, so that $g(y) = 2\log\cosh x$. The upper bound is $\cosh x \le e^{x^2/2}$, which holds term by term in the two power series. The lower bound says that $\chi(x) = \log\cosh x - \frac{x^2}{2} + \frac{x^4}{12}$ is nonnegative, and this holds as $\chi(0) = \chi'(0) = 0$ and $\chi'' = x^2 - \tanh^2 x \ge 0$. For (g2), $g'' = \frac{1}{2}\mathrm{sech}^2(y/2) > 0.2$ on $|y| \le 2$, so $g(y) \ge 0.1y^2$ there. For $|y| \ge 2$ convexity and $g(0) = 0$ give $g(y) \ge \frac{1}{2}|y|g(2) > 0.43|y|$.

Coordinates along and across the cyclic kernels. A cyclic kernel has $B_1 = 1 - B_2 = B_3$, so its logits are $\ell = tv$ with $v = (1, -1, 1)$ and $t \in \mathbb{R}$, and $M_1 = M_2 = M_3 = 0$ there. For general ℓ let $t = \langle \ell, v\rangle/3$. We use (t, M_1, M_2) as coordinates. The map $\ell \mapsto (t, M_1, M_2)$ is linear with determinant 1, and its inverse is $\ell = tv + w$ with $w = \frac{1}{3}(M_1 - M_2, 2M_1 + M_2, M_1 + 2M_2) \perp v$. So t measures the position along the line and w the displacement from it. Since $|v|^2 = 3$, $|\ell|^2 = 3t^2 + |w|^2$, and a direct computation gives $|w|^2 = \frac{1}{3}(M_1^2 + M_2^2 + M_3^2) = \frac{2}{3}(M_1^2 + M_1M_2 + M_2^2)$. Write $|M|$ and $|M|_\infty$ for the Euclidean norm and the maximum norm of (M_1, M_2). As $|M_1M_2| \le \frac{1}{2}|M|^2$,

$$\tfrac{1}{3}|M|^2 \le |w|^2 \le |M|^2 \le 2|M|_\infty^2. \tag{4}$$

Proof of Proposition 2.4(b). We compute the integral in (3) in four steps. Step 1 shows that on D the forms M_i are of second order in ℓ. Step 2 removes the part of D away from 0. Step 3 finds the section of D at t, and Step 4 integrates over t.

Step 1: an a priori bound. On D, each right side in (2) is at most ψ, so $M_1 \le \psi$, and $M_1 = -M_2 - M_3 \ge -2\psi$. The same holds for M_2. So by (g1), which gives $\psi \le \frac{1}{4}|\ell|^2$,

$$|M|_\infty \le 2\psi(\ell) \le \tfrac{1}{2}|\ell|^2 = \tfrac{1}{2}(3t^2 + |w|^2). \tag{5}$$

Step 2: localization. Put $\varepsilon = 1/12$ and let U be the set where $|t| \le \varepsilon$ and $|M|_\infty \le \varepsilon$. We show that the integral over $D \setminus U$ is exponentially small in a. Off U we have $|t| > \varepsilon$ or $|M| > \varepsilon$, so by (4) $|\ell| \ge \max(\sqrt{3}|t|, |M|/\sqrt{3}) \ge \varepsilon/\sqrt{3}$. Also some coordinate has $|\ell_j| \ge |\ell|/\sqrt{3}$. So (g2) gives

$$\psi \ge \gamma := \tfrac{1}{10}\min\big(|\ell|^2/3,\ |\ell|/\sqrt{3}\big) \ge \gamma_0 := \varepsilon^2/90.$$

Hence $a\psi \ge (a-1)\gamma_0 + \gamma$, that is, $e^{-a\psi} \le e^{-(a-1)\gamma_0}e^{-\gamma}$ off U. The function $e^{-\gamma(\ell)}$ is integrable over $\mathbb{R}^3$, so $\int_{D\setminus U} e^{-a\psi} = O(e^{-\gamma_0 a})$.

Step 3: the section of D at t. We show that the section of $D \cap U$ at t lies between two triangles of area close to $\frac{9}{8}t^4$. First, on $D \cap U$ the bounds (5) and (4) give

$$|M|_\infty \le \tfrac{3}{2}t^2 + \tfrac{1}{2}|w|^2 \le \tfrac{3}{2}t^2 + |M|_\infty^2 \le \tfrac{3}{2}t^2 + \varepsilon|M|_\infty,$$

so $|M|_\infty \le \frac{3}{2}t^2/(1-\varepsilon) \le 2t^2$. So $D \cap U$ lies in the set $E = \{|t| \le \varepsilon,\ |M|_\infty \le 2t^2\}$.

Next we estimate g_j and ψ on E. There $|w|^2 \le 8t^4$ by (4), and $\ell_j = \pm t + w_j$ with $|w_j| \le |w|$. So $|\ell_j| \le 2|t|$ and $|\ell_j^2 - t^2| \le 7|t|^3$. Then (g1) gives $|g_j - \frac{1}{4}t^2| \le 2|t|^3$, and with $|\ell|^2 = 3t^2 + |w|^2$ it bounds ψ. Hence on E

$$\big|g_i + g_j - \tfrac{1}{2}t^2\big| \le 4|t|^3 \quad (i \ne j), \qquad \big|\psi - \tfrac{3}{4}t^2\big| \le 2t^4. \tag{6}$$

So on E each right side of (2) is $\frac{1}{2}t^2$ up to $4|t|^3$. Now let $\mathcal{T}(\delta) = \{M : M_1, M_2 \le \delta,\ M_1 + M_2 \ge -\delta\}$. This is the set where $M_1, M_2, M_3 \le \delta$. It is a triangle with corners (δ, δ), $(\delta, -2\delta)$ and $(-2\delta, \delta)$, so it has area $\frac{9}{2}\delta^2$ and lies in $\{|M|_\infty \le 2\delta\}$. Let D_t be the section of $D \cap U$ at t and put $\delta_\pm = \frac{1}{2}t^2(1 \pm 8|t|) \ge 0$, which are $\frac{1}{2}t^2 \pm 4|t|^3$. If $M \in D_t$, then $(t, M) \in E$, and (2) and (6) give $M_1, M_2, M_3 \le \delta_+$. Conversely, if $M \in \mathcal{T}(\delta_-)$, then $|M|_\infty \le 2\delta_- \le \min(t^2, \varepsilon)$, so $(t, M) \in E \cap U$, and (6) gives (2). So $\mathcal{T}(\delta_-) \subseteq D_t \subseteq \mathcal{T}(\delta_+)$, and

$$\tfrac{9}{8}t^4(1 - 8|t|)^2 \le |D_t| \le \tfrac{9}{8}t^4(1 + 8|t|)^2. \tag{7}$$

Step 4: integration. Let $J = \int_{D \cap U} e^{-a\psi}$. We integrate first over the section and then over t (the change of coordinates has Jacobian 1). By (6) and (7), J lies between the integrals over $|t| \le \varepsilon$ of $\frac{9}{8}t^4(1 \mp 8|t|)^2 e^{-a(\frac{3}{4}t^2 \pm 2t^4)}$. We substitute $t = y/\sqrt{a}$. Then these are $a^{-5/2}$ times integrals over $|y| \le \varepsilon\sqrt{a}$ whose integrands tend to $\frac{9}{8}y^4 e^{-3y^2/4}$ as $a \to \infty$ and are at most $\frac{9}{2}y^4 e^{-y^2/2}$, since $8|t| < 1$ and $2y^4/a = 2t^2y^2 \le 2\varepsilon^2 y^2$. By dominated convergence and $\int_{\mathbb{R}} y^4 e^{-3y^2/4} dy = \frac{3}{4}\sqrt{\pi}(\frac{4}{3})^{5/2}$,

$$J = a^{-5/2}\big(\sqrt{3\pi} + o(1)\big).$$

By Step 2 the same holds for the integral over D. It remains to find Z. As Γ is log-convex, $\Gamma(a + \frac{1}{2})^2 \le a\Gamma(a)^2$ and $\Gamma(a+1)^2 \le (a + \frac{1}{2})\Gamma(a + \frac{1}{2})^2$, so $Z = 2\sqrt{\pi/a}\,(1 + O(1/a))$. Then (3) gives

$$\mathbb{P}_\omega(D) = \frac{a^{3/2}}{8\pi^{3/2}} \cdot a^{-5/2}\sqrt{3\pi}\,(1 + o(1)) = \frac{\sqrt{3}}{8\pi a}(1 + o(1)) = \frac{3\sqrt{3}}{8\pi\omega}(1 + o(1)). \qquad \square$$

Computations

The checks of this note are in the script `verify_sticky_fisher.py` in the folder `kagey-problems/problem131/paperC` of https://github.com/apemm/Kagey-Problems (its part on sticky priors), and the finite steps are in the Lean module `lean/PaperC/Sticky.lean`.

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